

Callum J. Hodge

07951 602615 | callum.hodge6@gmail.com | Coventry, UK | callum-j-hodge.github.io/my-website/

EDUCATION:

University of Warwick Coventry, UK

(Sep. 2023 - June 2027)

MPhys Physics with Astrophysics

Current Weighted Average mark: 91% (1st)

Top Module Marks

Electrodynamics:	99%	Computational Physics:	98%
Black Holes, White Dwarfs & Neutron Stars:	96%	Mathematics for Physicists:	96%
Quantum Physics of Atoms:	96%	Astrophysics Laboratory II and Skills:	85%

PUBLICATIONS & RESEARCH EXPERIENCE:

First Author on Research Paper submitted to *Monthly Notices of the Royal Astronomical Society* (June – Aug. 2026)

- 8-week research project determining the first estimate for the spin-down rate of a special white dwarf, in one of only three known white dwarf pulsars (a type of binary system that undergoes periodic, pulsed emission).
- Funded by winning a Royal Astronomical Society undergraduate research grant.
- Full research paper: (currently submitted and under peer-review).

Summer Research with the *University of Warwick Undergraduate Research Support Scheme* (June – Aug. 2025)

- 8-week research project studying a phenomenon known as the *Brown Dwarf Desert*, where brown dwarfs orbiting main sequence stars are largely observed at wide separations. Project involved python simulations.
- Funded by winning an undergraduate research grant from the University of Warwick.
- Full research paper: <https://urss.warwick.ac.uk/items/show/806>.

Conference Presenting at the *British and International Conferences of Undergraduate Research* (April & Sep. 2026)

- Presenting my own astrophysics research in poster and presentation formats to other undergraduate students from across the country for a total of four days across both conferences.
- Astrophysics research presented was from both projects listed above.

Final-year Undergraduate Research Project (Ongoing) (Sep. 2026 – Present)

- Using radio observations to track the blast-wave of a supernova. This gives us insight into the properties of the surrounding medium and allows us to track the history of mass-loss of the stellar progenitor.

AWARDS & HONOURS:

Physics Distinction List for Third Year

(Sep. 2025 – July 2026)

- Awarded to the top 10% of each cohort every year. Recognition of academic excellence in physics.

Duke of Edinburgh Award (Bronze and Gold)

(Sep. 2019 – Aug. 2022)

- National youth development award that builds skills via volunteering, physical activity and skills development.

Warwick Award: Gold

(Sep. 2023 – Sep. 2024)

- Awarded for proficiency in core employability skills from 300 hours of academic and extra-curricular activities.

- Skills include critical thinking, organizational awareness, problem solving, teamwork and professionalism.

COMPUTING & LABORATORY SKILLS:

- High proficiency in Python, C and LaTeX. Average mark of 94% across three years of Python/C coding modules.
- Analysed high-time resolution photometric data from ULTRACAM for my project on white dwarf pulsars (above).
- Extensive, hands-on experience using equipment such as telescopes, spectrographs and electronics from lab modules. Average mark of 79% across three years of lab modules.
- Experience using astrophysics software such as SAO ds9, Aperture Photometry Tool, VIREO and FireCapture.

SERVICE & OUTREACH:

Revision Lectures

(May 2024 – May 2026)

- Delivery of nine hours' worth of revision lectures to over 100 other undergraduate students.
- Content consisted of that from the modules *Classical Mechanics & Special Relativity*, *Quantum Mechanics & Its Applications*, *Stars & The Solar System* and *Black Holes, White Dwarfs & Neutron Stars*.

Volunteering with the University of Warwick Mobile Planetarium

(June 2026 – Present)

- Delivering educational shows to schools and youth organisations on astronomy, space exploration and the importance of science.

Live Labs and Resonate Roadshow

(Dec. 2023 & Dec. 2025)

- Volunteering at events held to encourage local people and school children to immerse themselves in science.
- Running workshops that create visual experiences for the general public on different topics in astrophysics.

University of Warwick Department of Physics Student Ambassador

(Sep. 2021 – Aug. 2023)

- Working on University open days to give tours of the campus, deliver information talks and answer questions.
- Communicating and presenting for tours and talks, managing groups of prospective students, enacting professionalism and key decision-making when answering questions and giving advice.

Maths Mentoring

(Jan. – May 2024)

- Volunteering as a mentor for two students taking GCSE maths.
- Organising and directing weekly revision sessions, as well as writing and marking weekly homework sheets.

EXTRA-CURRICULAR ACTIVITIES:

Warwick University Formula 1 Society Executive

(April 2024 – April 2026)

President, Treasurer and Content Creation Officer

- Role involves organising society events, leading society executive team, managing finances and social platforms.
- Delegating tasks to other executive members, balancing this workload with my degree, communicating and collaborating with external agencies, being a team player in my society executive team.

TeamWork 2024 International Experience Programme

(June – July 2024)

- Participation in 4-week research project with overseas students, working for McDermott International. Project aimed to evaluate different types of renewable energy sources to advise the company on future investment.
- Taking leadership over a project, maintaining punctuality to work towards deadlines, collaborating and communicating with members of my team effectively, dealing with complex issues in our research.

PART-TIME EMPLOYMENT:

Co-op Dartmouth, UK

(Aug. 2023 – Nov. 2024)

Shop Assistant

- Management and organisation of main store, serving and communicating with customers on till.
- Relevant skills include a logical approach to solving issues in main store, communication and presentation with customers on till, ability to collaborate with other employees, balancing and prioritising my workload.

John Fowler Holiday Park Brixham, UK

(Mar. 2022 – July 2023)

Sous Chef

- Waiting tables, taking orders on a till, communicating with customers, preparing dishes.
- Relevant skills include ability to work alone and using my initiative to solve customer issues, ability to work well with other employees towards a shared goal, attention to detail in organisation of workplace.

Venus Cafe Broadsands, UK

(June 2021 – Feb. 2022)

Front of House

- Taking orders on a till, preparing customer orders, warehouse management.
- Relevant skills include ability to work under time pressure in preparing orders, coherent communication with staff and customers, punctuality in meeting store deadlines. – not sure if this section is now relevant

INTERESTS:

- I volunteered as a mentor for students completing their maths GCSE course, in which I directed weekly online revision sessions.
- I have run multiple workshops at the University of Warwick's *Live Labs* and *Resonate Roadshow*, events held to encourage local people and school children to immerse themselves in science.
- In my spare time, I am also a keen competitive swimmer and squash player. – not sure if I need this

Churston Ferrers Grammar School Paignton, UK

(Sep. 2016 – June 2023)

GCSE and A-Level Qualifications

- Grade A* in all three A-levels of maths, physics and psychology.
- Grades 8-9 in nine GCSEs including maths, English, physics, biology and chemistry. – not sure if this is relevant